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POSITION PAPER

Diagnosis of Quinolone Hypersensitivity: An EAACI Position Paper

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ABSTRACT

Hypersensitivity reactions to quinolones (QHRs) have been increasing in frequency, thus classifying them as the second most frequently implicated class of antibiotics in hypersensitivity reactions (HRs). It is noteworthy that quinolones (Qs) have been observed to predominantly trigger immediate hypersensitivity reactions (IHRs). These reactions are categorized as either IgE-mediated or non-IgE-mediated, attributable to the off-target occupation of the recently described receptor, Mas-related G-protein coupled receptor member X2 (MRGPRX2), on effector cells. The increasing trend of HRs underscores the necessity for enhanced diagnostic and management strategies. The present position paper aims to shed light on the key mechanisms involved in immediate and non-immediate QHRs. The clinical spectrum of these reactions is discussed, as well as the utility of skin tests, in vitro diagnostic tests, and drug provocation tests in diagnosis. A further focal point of this study is the analysis of cross-reactivity between various Qs. The paper concludes with the presentation of diagnostic algorithms for both immediate and non-immediate QHRs. The paper's findings aid clinical practice for QHRs and address unmet needs, which should stimulate more in-depth investigations into the mechanisms and clinical practice of QHRs.

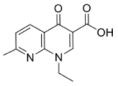
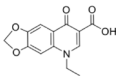
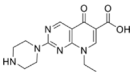
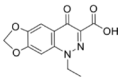
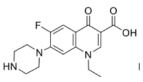
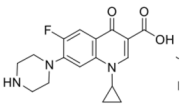
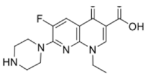
1 | Introduction

Quinolones (Qs) are classified into four generations based on their chemical structure, pharmacokinetic properties, and antibacterial activity (Table 1). Fluoroquinolones (FQs) possess a fluorine atom at the position C-6 and form the second, third, and fourth generations [1, 2].

Qs are generally well-tolerated [3]. However, the prevalence of hypersensitivity reactions (HRs) to these drugs has been

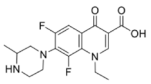
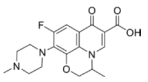
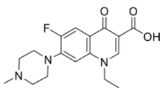
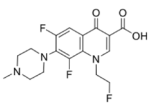
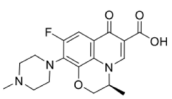
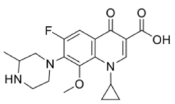
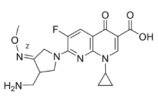
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TABLE 1 | Main characteristics of quinolones [1].

Type	Generation	Drug name	Chemical structure	Mechanism of action	Elimination route	Antibacterial spectrum
Quinolone	1st generation	Nalidixic acid (first discovered quinolone)		Inhibition of DNA synthesis of bacteria by interfering the activity of topoisomerase type II (DNA gyrase) and IV	Renal	Gram-negative bacteria (Except *Pseudomonas* spp.)
		Oxalinic acid			Renal	
		Pipemidic acid			Renal	
		Cinoxacin			Renal and other	
Fluoroquinolone	2nd generation	Norfloxacin			Renal and hepatic	Gram-negative bacteria *Mycoplasma pneumoniae* *Chlamydia pneumoniae*
		Ciprofloxacin			Renal and enteral	
		Enoxacin			Renal	

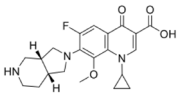
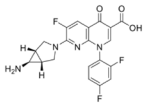
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TABLE 1 | (Continued)

Type	Generation	Drug name	Chemical structure	Mechanism of action	Elimination route	Antibacterial spectrum
		Lomefloxacin			Renal	
		Ofloxacin (L-isomer form of Levofloxacin)			Renal	Gram-negative bacteria Gram-positive bacteria (except *Streptococcus pneumonia*) Some atypical microorganisms (*Mycoplasma pneumoniae* and *Chlamydia pneumoniae*)
		Pefloxacin			Hepatic	
		Fleroxacin			Renal and other	
	3rd generation	Levofloxacin			Renal	In addition to activities of 2nd generations, improved activity against Gram-positive and atypical pathogens
		Gatifloxacin			Renal	
	4th generation	Gemifloxacin			Renal and other	In addition to activities of 3rd generation drugs, anaerobic activity

(Continues)

TABLE 1 | (Continued)

Type	Generation	Drug name	Chemical structure	Mechanism of action	Elimination route	Antibacterial spectrum
		Moxifloxacin			Hepatic	
		Trovafloxacin			Hepatic	

increasing, most likely due to their expanded use in recent decades and the introduction of structurally different Qs [2, 4–9]. Qs, as the third most commonly used drugs [5, 7] and the second most frequently applied antibiotics associated with HRs [6], induce predominantly immediate HRs (IHRs) [4, 6, 7] and less frequently non-immediate HRs (NIHRs) [2, 6, 7]. IHRs manifest within 6 h of Q administration and are classified as either IgE-mediated or non-IgE mediated via off-target occupation of the recently described receptor, the Mas-related G-protein coupled receptor member X2 (MRPRX2) on effector cells [mast cells (MCs) or basophils] [2, 4, 10–18]. NIHRs may occur from 1 h to about 2 weeks after initial drug administration and involve T effector cells [2, 10, 19].

Clinical suspicion of hypersensitivity reactions to Qs (QHRs) is based on a pattern of suggestive manifestations and their onset in relation to the administration of the suspected offending agent. However, as shown in Figure 1, similar clinical pictures can be caused by different mechanisms, and differential diagnoses, such as underlying infections, have to be considered. The diagnosis of QHR may be challenging because of an unreliable clinical history [4, 20], unknown pathomechanism, limitations in the sensitivity and specificity of skin tests (STs) [11, 20–22] and in vitro tests [11, 20–23], and may often require drug provocation test (DPT) [11, 20–22, 24].

The aim of this position paper is to present the current knowledge on the pathomechanism of QHRs, diagnostic tests and cross-reactivities, and to develop a practical diagnostic approach for QHRs for clinicians in the field.

2 | Methods

This position paper was commissioned by the European Academy of Allergy and Clinical Immunology (EAACI). During the development of this position paper, online meetings were held in April 2021 and September 2021, and a final revision was performed via emails in January 2025. Members of this Task Force Group reviewed the literature in PubMed on QHRs, focusing on the words/terms FQ/Q hypersensitivity, Q hypersensitivity, Q allergy, the name of each Q (ciprofloxacin, levofloxacin, moxifloxacin, norfloxacin, nalidixic acid, and pipemidic acid), anaphylaxis, urticaria, exanthema, IHRs, NIHRs, delayed reactions, ST, skin prick test (SPT), intradermal test (IDT), in vitro test, basophil activation test (BAT), lymphocyte transformation test (LTT), DPT, and MRGPRX2.

3 | Epidemiology

FQs, mainly moxifloxacin, ciprofloxacin, and levofloxacin, are the most common non-beta-lactam antibiotics involved in HRs [4, 5, 7], also among hospitalized patients [25]. In the USA, the absolute risk of a QHR of any severity has been reported to be low [44.0 (95% CI: 34.8–53.3) emergency department (ED) visits/100,000 prescriptions] [26, 27]. However, epidemiological data on QHRs suggest an increase in the incidence of QHRs [2, 4, 5, 7]; for example, in Spain from 0.54% in 2005 to 6.85% in 2010 [5].

Qs are reported to be responsible for 4.5% of cases with drug-induced anaphylaxis. The incidence of Q-induced anaphylaxis is 4% in hospitalized patients [27] and is estimated to be 1.8–2.3 per 10 million treatment days [28, 29]. Although less common than IHRs, NIHRs have also been reported [19, 30–34]. In a recent study by Doña et al. of 128 patients with a confirmed diagnosis of QHRs, 18 (26.2%) had NIHRs [6].

4 | Risk Factors

Data on risk factors for the development of QHRs are limited. The culprit Qs involved in HRs influences the risk of anaphylaxis. In Germany; analysis of 166 cases showed that moxifloxacin, levofloxacin, ciprofloxacin and ofloxacin were responsible for 54%, 15%, 13% and 10% of anaphylaxis caused by Qs, respectively [35]. Similarly, in the USA, the relative risk of ED visits due to HRs was higher for moxifloxacin [141.3 (95% CI: 99.9–182.7)] than for other Qs including levofloxacin [40.8 (95% CI: 31.5–50.0)] and ciprofloxacin [26.3 (95% CI: 20.8–31.9)] [26]. Consequently, moxifloxacin appears to be the most common cause of anaphylaxis induced by Qs [26, 35]. "A confirmed beta-lactam allergy has been linked to a significantly increased risk of QHRs. In one study, patients with a beta-lactam allergy verified by DPT had approximately a 23-fold higher risk of developing QHRs compared to those without such an allergy" [4].

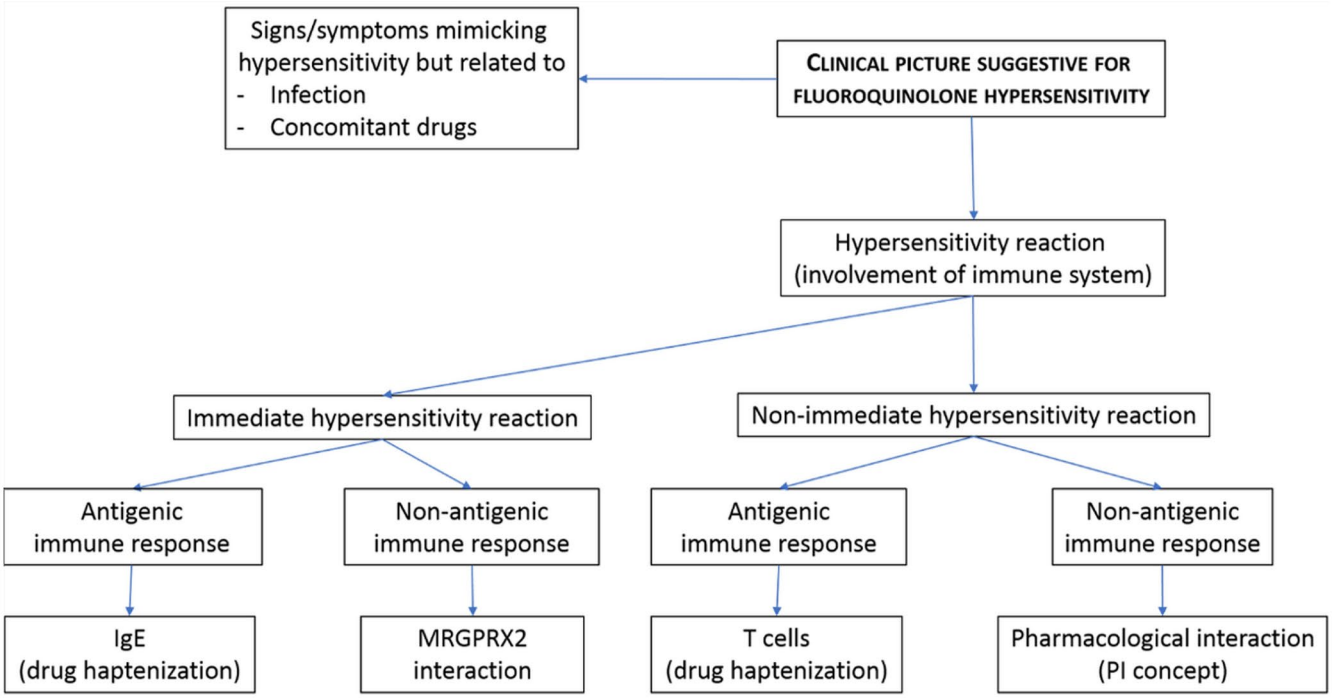


FIGURE 1 | Pathogenesis of quinolone-induced hypersensitivity reactions.

For NIHR, ciprofloxacin was reported to be the most common culprit Q, followed by levofloxacin and moxifloxacin [9, 36], and was also associated with a higher risk of developing severe cutaneous adverse drug reactions (SCARs) [37]. No risk factors for non-immediate QHRs have been identified so far. One recent Iranian study reported a possible association between ciprofloxacin-induced delayed urticaria to HLA-B*40 and HLA-DRB1*14 and in acute generalized exanthematous pustulosis (AGEP) to HLA-B*08 in a few patients [38]. A recent study conducted in the Han Chinese population demonstrated that the carrier frequencies of HLA-B*13:01, HLA-B*13:02, and serotype B13 in patients with levofloxacin-induced SCARs were significantly higher than those of levofloxacin-tolerant individuals [39]. However, there are no studies available in other populations yet.

5 | Pathomechanism

Chronological and morphological features that characterize IHRs and NIHRs imply specific pathological activation of immune system cells and/or inflammatory mechanisms (Figure 1). The involvement of the adaptive immune system triggers a so-called antigenic immune response. On the other hand, in non-antigenic immune responses, reactions do not involve the adaptive immune system [18, 40, 41].

5.1 | Immediate Antigenic Immune Response

In IHRs, drug-specific IgE (sIgE) antibodies may specifically activate the adaptive immune system. They bind to high-affinity receptors (FcεRI) on the effector cells of IHRs, i.e., MCs and basophils. Re-exposure to the sensitizing drug (or a structurally similar compound), which cross-links these sIgE:FcεRI complexes, triggers degranulation of these effector cells with the release of a variety of mediators responsible for the clinical phenotype of IHR [42]. To date, in vitro studies aimed at determining sIgEs for Qs have provided conflicting results with variable sensitivity and specificity [23, 43].

5.2 | Immediate Non-Antigenic Immune Response

Since the seminal description by McNeil et al. in 2015 [12], an increasing number of studies (for review: [18, 44]) have shown that Qs can directly trigger nonspecific MC activation (MCA) and degranulation by occupying the MRGPRX2. However, it should be noted that our knowledge of MRGPRX2 signaling by Qs is based solely on preclinical studies. Alternatively, this mechanism could explain why many patients with an immediate QHR are drug-naïve. This mechanism also explains the uncertainties with Q skin testing [45] and the low sensitivity of BATs [15]. Resting basophils, which are used as a starting point for BATs, barely express MRGPRX2, in contrast to skin MCs. Furthermore, although not definitive, it is noted that patients who reported an IHR were more likely to develop anaphylaxis that was specific to a particular Q and did not have multiple drug HRs that included other MRGPRX2 agonists such as opiates, vancomycin, and curarizing neuromuscular blocking agents (NMBAs) [46].

5.3 | Non-Immediate Reactions

Q can also trigger NIHRs, but less commonly than IHRs. Their pathogenesis is thought to be T cell-dependent. Schmid et al.

TABLE 2 | Drug concentrations recommended for skin tests to quinolones in immediate reactions.

References	Year	Drug	SPT	IDT	Comments[a]
Dávila et al. [60]	1993	Nalidic acid	400 mg/mL	400 mg/mL	Negative in two patients
		Ciprofloxacin	2 mg/mL	ND	Positive in a patient, but also in "several" controls
Erdem et al. [61]	1999	Ciprofloxacin	2 mg/mL	0.004 mg/mL	Number of controls not specified
Empredrad et al. [57]	2003	Levofloxacin	ND	0.025 mg/mL	Nonirritating concentration determined in 25 control subjects
		Ciprofloxacin	None	None	At least 1/25 patients positive to high dilution, authors do not recommend skin tests
González et al. [62]	2005	Ciprofloxacin	0.02 mg/mL	0.02 mg/mL	Non-irritant in 10 controls
		Ofloxacin	Tablet suspended in NaCl	ND	
		Levofloxacin	5 mg/mL	0.05 mg/mL	
		Norfloxacin	Tablet suspended in NaCl	ND	
		Moxifloxacin	Tablet suspended in NaCl	ND	
Scherer et al. [24]	2005	Ciprofloxacin	ND	0.002 mg/mL	Non-irritant in nine controls
Venturini Diaz et al. [65]	2007	Ciprofloxacin	0.02 mg/mL	0.02 mg/mL	Authors interpret wheals in IDT > 5 mm after 20 min as positive, 63% of 12 patients positive to at least one skin test, 12 controls had negative ST to CFX, LFX, NFX, but positive to MFX ($n = 5$), OFX ($n = 3$), and PA ($n = 1$)
		Ofloxacin	Tablet suspended in NaCl	NP	
		Levofloxacin	5 mg/mL	0.05 mg/mL	
		Norfloxacin	Tablet suspended in NaCl	ND	
		Moxifloxacin	Tablet suspended in NaCl	ND	
		Pipemidic acid	Tablet suspended in NaCL	ND	
		Trovafloxacin	Tablet suspended in NaCL	ND	
Sánchez–Morillas et al. [63]	2010	Moxifloxacin	4 mg/mL	0.004 mg/mL	SPT negative in one patient, IDT 0.04 mg/mL positive in 10 controls and patient, and IDT 0.004 mg/mL positive in the patient
Seitz et al. [20]	2009	Ciprofloxacin	2 mg/mL	ND	Authors find skin test not very helpful, as SPT were positive in 12/64 patients and positive in provocation test in 3/6 (50%) challenged patients. Authors found no reliable concentration to perform IDT
		Ofloxacin	2 mg/mL	ND	
		Levofloxacin	5 mg/mL	ND	
		Moxifloxacin	1.6 mg/mL	ND	
Chang et al. [64]	2010	Ciprofloxacin	ND	0.05 mg/mL	Negative in two patients, no controls
		Levofloxacin	ND	0.05 mg/mL	Negative in two patients, no controls
		Moxifloxacin	ND	0.05 mg/mL	Negative in one patient, no controls

(Continues)

TABLE 2 | (Continued)

References	Year	Drug	SPT	IDT	Comments[a]
Ben Said et al. [21]	2010	Ciprofloxacin	5 mg/mL	0.005 mg/mL	Negative in five controls
		Ofloxacin	5 mg/mL	0.005 mg/mL	Negative in five controls, positive SPT in two patients
		Levofloxacin	5 mg/mL	0.005 mg/mL	Negative in five controls, positive SPT and IDT in one patient each
		Moxifloxacin	5 mg/mL	0.005 mg/mL	Negative in five controls, positive SPT in one patient
Lobera et al. [22]	2010	Ciprofloxacin	0.02 mg/mL	0.02 mg/mL	Skin tests positive in 5/12 patients (2 LFX, 2 MFX, 2 OFX), no reactions in 12 controls
		Ofloxacin	Tablet suspended in NaCl	ND	
		Levofloxacin	5 mg/mL	0.05 mg/mL	
		Norfloxacin	Tablet suspended in NaCl	ND	
		Moxifloxacin	Tablet suspended in NaCl	ND	
Broz et al. [66]	2012	Ciprofloxacin		0.0067 mg/mL	Not irritant in 15 controls
Perez et al. [58]	2013	Ciprofloxacin	2 mg/mL	0.02 mg/mL	Authors do not recommend skin tests as only 3/27 patients with positive skin tests to all FQs were positive in the provocation test and 3/4 patients with a positive provocation test had a positive skin test
		Ofloxacin	5 mg/mL	0.05 mg/mL	
		Levofloxacin	5 mg/mL	0.05 mg/mL	
		Norfloxacin	Tablet suspended in NaCl	ND	
		Moxifloxacin	Tablet suspended in NaCl	ND	
Rouzaire et al. [56]	2013	Ofloxacin	5 mg/mL	0.05 mg/mL	Additionally 1/10 of IDT concentrations, French positivity criteria, 12/17 patients positive to their culprit FQ, 1/9 DPT-negative controls had positive ST to levofloxacin, the others negative to their suspected but not confirmed FQ
		Levofloxacin	5 mg/mL	0.05 mg/mL	
		Norfloxacin	40 mg/mL	0.4 mg/mL	
		Moxifloxacin	40 mg/mL	0.4 mg/mL	
		Pipemidic acid	40 mg/mL	0.4 mg/mL	
		Lomefloxacin	40 mg/mL	0.4 mg/mL	
Uyttebroek et al. [67]	2015	Moxifloxacin	1.6 mg/mL	0.0016 mg/mL	IDT also at 0.0016 + 0.16 mg/mL; SPT negative in all patients, positive MFX IDT in 10/14 MFX allergic patients (2 at 0.0016 mg/mL; 2 at 0.016 mg/mL; 6 at 0.16 mg/mL) and in 12/16 MFX tolerant controls tested (2 at 0.016 mg/mL; 12 at 0.16 mg/mL). Authors consider skin tests as unreliable
Demir et al. [68]	2017	Ciprofloxacin	5 mg/mL	0.05 mg/mL	11/57 skin tests positive to culprits (5 CFX, 6 MFX, 1 LFX); 18/46 patients developed reaction despite negative skin test; results with alternative FQ diverse, overall sensitivity of 3.8%, specificity of 95.2%, authors conclude that skin tests are not useful to indicate alternatives for all drugs and are useful to confirm diagnosis only for CFX, MFX, LFX
		Ofloxacin	4 mg/mL	ND	
		Levofloxacin	5 mg/mL	0.05 mg/mL	
		Moxifloxacin	4 mg/mL	0.04 mg/mL	
		Gemifloxacin	3.2 mg/mL	ND	

(Continues)

TABLE 2 | (Continued)

References	Year	Drug	SPT	IDT	Comments[a]
Krantz et al. [45]	2021	Ciprofloxacin	ND	0.025 mg/mL	Substances also tested at 0.005 mg/mL. Different positivity criteria for IDT were compared. Authors propose criterium for positive IDT: specific FQ flare at 0.025 mg/mL ≥ histamine flare, specific FQ flare ≥ 5 mm at 0.005 mg/mL, and no flare ≥ 5 mm for either of the other 2 FQ at 0.005 mg/mL to discriminate the few IgE-mediated reactions from non-immunological reactions with only 2% positivity in cohort
		Levofloxacin	ND	0.025 mg/mL	
		Moxifloxacin	ND	0.025 mg/mL	
Chiriac et al. [69]	2021	Ofloxacin	5 mg/mL	0.05 mg/mL	Skin tests were positive in 9/100 patients (9% of the tested Patients, six to moxifloxacin) without the drug provocation test to the culprits
		Levofloxacin	1 mg/mL	0.01 mg/mL	
		Norfloxacin	40 mg/mL	ND	
		Moxifloxacin	10 mg/mL	0.1 mg/mL	
		Pefloxacin	4 mg/mL	0.04 mg/mL	
		Ciprofloxacin	2 mg/mL	0.02 mg/mL	
		Lomefloxacin	40 mg/mL	ND	

Abbreviations: CFX, ciprofloxacin; FQ, fluoroquinolones; LFX, levofloxacin; MFX, moxifloxacin; ND, not done; NFX, norfloxacin; OFX, ofloxacin; PA, pipemidic acid.
[a]If the non-irritant test concentration was determined in the study, it is indicated in the "Comments column", and the maximum non-irritant test concentration is the same as the suggested test concentration listed in the SPT or IDT column of the table. This applies to the SPT column only if IDT was not performed in that study; definite non-irritant concentrations for individual quinolones have not been established yet.

demonstrated that NIHRs can elicit robust T cell immune responses against the relevant Q. They also provided evidence that T cells recognize the parent, chemically inert compound without the need for metabolism or processing. The authors suggested that NIHRs are induced by pharmacological interaction with immune receptors, or a "p–i" concept [19].

6 | Diagnosis of Immediate Hypersensitivity Reactions

6.1 | Clinical Presentation

The most common clinical presentations of immediate QHRs are urticaria with or without angioedema (26.4%–85%) and anaphylaxis (19.3%–53.8%) [4, 5, 7, 47]. For moxifloxacin, the most common manifestation is anaphylaxis (70%) [6]. Q-induced anaphylaxis can be severe, leading to myocardial infarction (Kounis syndrome), which is mainly associated with ciprofloxacin, levofloxacin, and gemifloxacin [48–51]. Biphasic anaphylaxis due to gemifloxacin has been reported [52]. In addition, a case report describes a dual DHR due to a single dose of moxifloxacin during DPT in a patient with multidrug hypersensitivity, involving anaphylaxis during DPT and a maculopapular eruption (MPE) that developed 24 h after recovery from anaphylaxis [53].

6.2 | Skin Tests

The value of using STs in IHRs is yet to be proven due to a high rate of false positive ST reactions, very low positive predictive value, lack of validated ST concentrations, and unknown sensitivity [2, 54, 55]. STs may be confounded by the fact that Qs can cause nonspecific MC degranulation, even in many controls, possibly through the histamine-releasing action of the MRGPRX2 pathway [2, 17, 20].

Several authors have investigated ST concentrations in an attempt to find non-irritative concentrations (NICs) with acceptable sensitivity. However, unfortunately, results vary between studies [2, 21, 24, 56–64]. Several authors found STs useful for the evaluation of Q IHR. However, there is a manifold difference in recommended concentrations. For ciprofloxacin, levofloxacin, moxifloxacin, norfloxacin, and ofloxacin, some recommend SPT concentrations in the range of 2–5 mg/mL, and about 1/100 of that concentration for IDT [2, 20, 24, 58, 65]. Other authors found that much lower concentrations, as low as 0.025 mg/mL depending on the Q, can be non-irritative in SPT [57, 66]. Therefore, the optimal ST concentration for each individual molecule remains unknown, and no firm recommendations can be made. Some experts advocate SPT alone, while others use IDTs, and some do not advocate skin testing with Qs at all [2, 22, 47, 67, 68]. The study with the largest number of patients found some value in performing SPT [20]. Of 64 patients, who all had IHR to Qs and had SPTs with moxifloxacin (1.6 mg/mL) and levofloxacin (5 mg/mL), 12 tested positive and 52 tested negative. In the DPT, whereas 50% of the 6 patients with a positive SPT developed urticaria, only 7% of the 52 patients with a negative SPT reacted positively. A study of patients with IHR to moxifloxacin IDT found overlapping results in patients and exposed controls [67]. Two of 14 patients tested positive at a dilution of 1:1000, whereas none of the 16 controls tested positive. At a 1:100 dilution,

2 of 14 patients and 2 of 16 controls tested positive. At a dilution of 1:10, 6 of 14 patients and 12 of 16 controls tested positive (Table 2).

In conclusion, ST may be helpful when a Q is suspected to be the cause of an IHR, if results are cautiously interpreted and if positive results occur to higher dilutions. Negative ST results should always be followed by DPT to exclude IHR, whereas positive results should be followed up by either avoidance or a DPT of the clinical picture and the clinician's suspicion. Without clinical suspicion of Q allergy, the sensitivity and specificity are not reliable enough to recommend STs. Due to the high rate of false positive reactions, it is strongly advised against STs as a screening test for Q IHR.

Further research is required on this topic with an open mind. For instance, it is necessary to strictly define the ST procedure, including the exact dilution and volume used for IDT [70]. Criteria for positive ST reactions to each individual Q, as has been done for moxifloxacin [45], should also be defined. NICs in tolerant controls must be defined before performing and interpreting STs in the clinical routine outside of the setting of allergists experienced with drug hypersensitivity.

6.3 | In Vitro Tests

For in vitro diagnosis of immediate QHRs, there is a controversy about the most useful approach due to the possible different mechanisms of MCA either Q-sIgE mediated or off-target interactions with MRGPRX2.

For sIgE-mediated reactions, immunoassays and more recently flow cytometric-based BAT have been used mostly with in-house methods [11, 23, 71, 72]. In the case of immunoassays, in-house sepharose-radioimmunoassays were used in only two studies which showed a low sensitivity (31.6%–54.5%) and high specificity [11, 23] (Table 3). Differences in sensitivity could be due to the Q involved and the severity of the reaction [78]. This test is currently not commonly used due to its methodological difficulties and low sensitivity.

BAT has been increasingly used in recent years to assess these IgE-mediated reactions and has also shown discrepant results with sensitivities ranging from 0% to 89.5% [6, 11, 15, 68, 73, 75, 79]. In a prospective study, BAT was positive in 68 (89.5%) cases, mainly to ciprofloxacin in 80% and to a lesser extent to moxifloxacin (62.5%) and levofloxacin (45.4%) with a high cross-reactivity between these Qs [6]. Importantly, this technique has a good negative predictive value and may help to decide whether to perform DPT in suspected QHR [79]. Specificity ranges from 77.8% to 100% [6, 11, 15, 22, 68, 73–75, 79] although the possibility of not detecting IgE-independent pathways must be considered. Differences in sensitivity may depend on the responsible Q and the activation marker used. In fact, lower sensitivity is obtained using CD63 expression than with upregulation of CD203c, although this may depend on the Q involved since CD63 upregulation occurs mainly when ciprofloxacin is included in the test, whereas for moxifloxacin, the CD203c could be used, although with suboptimal results [15, 74, 80] (Table 3).

In a recent study, BAT was positive for the culprit in only two cases; interestingly, only CD63 was used to assess activation [68]. All these results suggest that, at least for Q-IgE mediated reactions, the dual labeling staining for CD63 and CD203c could help increase BAT global sensitivity. Moreover, other factors like other Q inclusion [11] or possible photodegradation avoidance [73] could also improve BAT sensitivity.

Given the increasing evidence that IHRs to Qs may result from MRGPRX2 off-target occupancy [17, 81], this mechanism should be evaluated [12] as well as the real clinical relevance. However, there is currently no validated assay to test MRGPRX2-dependent reactions; in fact, traditional BAT uses resting basophils, which barely express MRGPRX2 as a starting point [82, 83], making them unsuited to capture MRGPRX2 signaling [13, 15, 68, 81]. According to the preclinical data, Qs activate MRGPRX2 in a dose-dependent manner and can induce mast cell degranulation on first exposure in naïve models [81, 82].

While recent developments in mast cell activation testing (MAT) show encouraging diagnostic potential, no published studies to date have documented the direct use of MAT specifically involving Q compounds.

6.4 | Drug Provocation Tests

In IHR to Qs, DPT with the culprit drug is indicated for those with questionable history and in those with a positive history, but negative allergy assessment (in vivo and/or in vitro tests) and mild initial reactions, such as urticaria and angioedema. DPT with alternative Qs may be indicated in cases with positive DPT or a positive allergy assessment with the culprit Q, and in severe initial reactions such as anaphylaxis even despite a negative allergy assessment of the culprit drug [2].

DPT is a procedure that carries risks [2]. Therefore, it is important to discuss the risks, benefits, and potential alternatives of DPT with the patient and obtain their signed consent before proceeding [84]. There is currently no consensus on the number of steps, time between steps, and dose in the protocols. Most challenge protocols published for immediate QHRs are multistep protocols. Most protocols start with administering 1/100th of the total dose, with 60-min intervals, and gradually increase the dose until the full dose is reached in 1 day [4, 6, 20, 22, 62, 64, 65, 68, 69, 75, 79]. However, some publications suggest a two-step protocol, which involves administering 1/10th and 9/10th of the dose. After an uneventful multistep challenge, there is no need to avoid future courses of the drug [84, 85]. In case of positive results, an allergy passport has to be issued and the patient has to be informed to avoid all Qs not cleared by DPT [86].

Topical Qs are widely used as eye or nasal drops, which can cause IHRs that are sometimes underrecognized, leading to a systemic reaction after a subsequent exposure [87–89]. Therefore, routes of administration other than oral, such as intravenous or topical, could also be considered for the DPT.

Patients with IgE-mediated reactions could lose IgE antibodies without re-exposure as happens with betalactams [11, 23, 90]. Therefore, the interval between the initial reaction and the DPT

TABLE 3 | In vitro test results of immediate and non-immediate hypersensitivity reactions to quinolones.

Paper	*N* Pat	Responsible quinolones	Clinical symptoms	Diagnostic tests	In vitro method	Activation marker	Sensitivity (%)	Specificity (%)
Manfredi et al. [23]	55	Quinolones	Anap S 13% Anap 35% Urt 85%	CH	Sepharose-RIA		54.5	100
Aranda et al. [11]	38	Cip, Mox, Lev	Anap S 26.3% Anap 42.1% Urt 31.6%	CH, DPT	Sepharose-RIA		31.6	100
Seitz et al. [20]	4	Cip, Mox, Lev	Anap	CH, DPT	BAT	CD63	0	ND
Lobera et al. [22]	12	Cip, Mox, Lev	Anap 33% Urt/Ang 66%	CH, ST, DPT	BAT	CD63	0	100
Ben Said et al. [21]	5	Mox, Lev, Oflo	Anap	CH, ST	BAT	CD203c	100	100
Aranda et al. [11]	38	Cip, Mox, Lev	Anap S 26.3% Anap 42.1% Urt 31.6%	CH, DPT	BAT	CD63	71	88
Rouzaire et al. [56]	34	Quinolones	Grade 1 32% Grade 2 21% Grade 3 47%	CH, ST, DPT	BAT	CD203c	50	ND
Mayorga et al. [73]	28	Cip, Mox	Anap S 11% Anap 64% Urt 25%	CH, DPT	BAT	CD63	Light 46.4 Dark 57.1	Light 90 Dark 90
Fernández et al. [74]	17	Cip, Mox	Anap S 29% Anap 41% Urt 29%	CH, DPT	BAT (Dark)	CD63 CD203c CD63/CD203c	52.9 23.5 64.7	72.3 88.9 66.7
	11	Mox	Anap S 45.5% Anap 36% Urt 18%			CD63 CD203c	9.1 36.4	77.8 94.4
	6	Cip	Anap 50% Urt 50%			CD63 CD203c	83.3 0	88.9 94.4
Van Gasse et al. [15]	15	Mox	Grade 2 66.6% Grade 3 33.3%	CH, DPT	BAT	CD63 CD203c	13.3 46.7	100 100
Demir et al. [68]	39	Cip, Mox, Oflo, Gemi	Anap 28% Urt/Ang 72%	CH, ST, DPT	BAT	CD63	23	ND

(Continues)

TABLE 3 | (Continued)

Paper	*N* Pat	Responsible quinolones	Clinical symptoms	Diagnostic tests	In vitro method	Activation marker	Sensitivity (%)	Specificity (%)
Doña et al. [6]	112	Quinolones	Anap 62.5% Urt/Ang 37.5%	CH, DPT	BAT	CD63	89.5	ND
	25	Cip					80	
	34	Mox					62.5	
	11	Lev					45.4	
Loli-Ausejo et al. [75]	19	Cip, Lev, Mox, Norflo	Anap 47% Urt/Ang 53%	ST, DPT	BAT	CD63	47	ND
Demir et al. [68]	13	Cip, Mox, Oflo, Gemi	Urt, Rash, FDE	CH, ST, DPT	LTT	CFSE	5	ND
Fernandez-Santamaría et al. [76]	4	Cip	SJS/TEN	CH, ST	LTT	CFSE	75	85
Rönnau et al. [77]	2	Cip	Urt/Ang (immediate)	CH, ST	LTT	^{3}H-Thymidine	100	ND
Scherer et al. [24]	10	Cip	MPE	CH	LTT	^{3}H-Thymidine	20	ND
Schmid et al. [19]	6	Cip, Mox	MPE, AGEP	CH, ST	LTT	^{3}H-Thymidine	100	100

Abbreviations: AGEP, acute generalized exanthematous pustulosis; Anap, anaphylaxis; Ang, angioedema; BAT, basophil activation test; CFSE, carboxyfluorescein succinimidyl ester; CH, clinical history; Cip, ciprofloxacin; DPT, drug provocation test; FDE, fixed drug eruption; Gemi, gemifloxacin; Lev, levofloxacin; LTT, lymphocyte transformation test; Mox, moxifloxacin; MPE, maculopapular exanthema; ND, not determined; Norflo, norfloxacin; Oflo, ofloxacin; RIA, radioimmunoassay; SCAR, severe cutaneous adverse reactions; SJS, Stevens-Johnson syndrome; ST, skin test; TEN, toxic epidermal necrolysis; Urt, urticaria.

should be taken into account to evaluate the need for repeat testing. Table 4 shows the protocol used in different studies and the rate of positive DPT with the culprit and alternative Qs.

7 | Non-Immediate Hypersensitivity Reactions

7.1 | Clinical Presentation

The most commonly reported NIHRs are delayed urticaria, MPE, being usually not severe, and fixed drug eruption (FDE), most commonly to ciprofloxacin [19, 30–34]. Although rare, cases of AGEP [91, 92], Stevens Johnson Syndrome (SJS), and toxic epidermal necrolysis (TEN) [32–34] have also been reported. Additionally, anecdotal cases have reported photosensitivity [93], vasculitis [94], hypersensitivity pneumonia [95], and drug rash with eosinophilia and systemic symptoms (DRESS) [96]. Risk factors for poor prognosis of SJS include older age, concomitant levothyroxine treatment, and HIV infection [37, 47]. Regarding AGEP, the EuroSCAR study found that Qs are the third most common drug associated with the condition, after pristinamycin and aminopenicillin [92]. Of the 97 validated cases of AGEP in the study, 9 were associated with Q intake, with the reaction occurring within 15 days of starting the drug in 7 of those cases. Ciprofloxacin, norfloxacin, and ofloxacin have been reported to carry a higher risk of AGEP compared to other Qs.

7.2 | Skin Tests

The value of patch testing and delayed readings of SPT and IDT for MPE from Qs remains low to unproven [59]. Patch testing has used 10%–20% Q in petrolatum or a tablet grounded and suspended in 1 mL saline. There is no evidence for false-positive reactions, but sensitivity appears to be poor. In a study with 37 patients with non-immediate QHR, (> 6 h after the last dose), none had a positive result [20]. However, 28 of these patients underwent Q graded challenges with the offending drug and only 2/28 patients reacted with exanthema after oral DPT. Thus, patch tests (PTs) of tablets ground in a mortar and suspended with 1 mL physiological saline solution had no value in 2/2 patients with proven NIHR. This needs to be confirmed in larger studies. On the other hand, positive PT results have been reported in individual cases of systemic contact dermatitis [97], in DRESS [98], and in AGEP [91]. It may be studied if delayed readings of IDT may have a better sensitivity than PTs for NIHR.

Photo-PTs have very low or negligible sensitivity in patients with phototoxic or photoallergic reactions. Furthermore, the results of PTs in the affected area and on the unaffected back are negative in most patients with FDE [30, 63, 99], with only a few exceptions [100, 101].

7.3 | In Vitro Tests

For non-immediate QHRs, studies using in vitro tests are scarce and most of them include few cases. However, most studies use LTT, which has demonstrated T cell involvement in the pathogenesis of Q-induced MPE and AGEP [19, 24, 102, 103]. As a result, it has been difficult to determine the sensitivity and specificity of the test in Q hypersensitivity. In two studies including 10 and 13 patients with non-immediate QHRs, the sensitivity of LTT was very low, at 20% [24] and 5% respectively [68]. The latter study mentioned the use of carboxy-fluorescein-succimidyl-ester (CFSE), and suggested that evaluation of other T cell populations may increase sensitivity, as has recently been demonstrated [76]. Additionally, clinical studies have shown cross-reactivity of T cells within the Q family, as noted in Table 3 and supported by references [19, 60, 77].

It has been observed that the immunological interaction is related to the chemically inert compound without the need of metabolism or processing [19]. However, recognition of Q metabolites cannot be completely excluded. Furthermore, this interaction with the immune system is labile, in contrast to other haptens such as sulfamethoxazole or penicillins [104]. It is possible that active intermediates may be formed that covalently bind to proteins/peptides acting as haptens [102].

Although lymphocyte responses to Qs have been demonstrated in patients with NIHRs using LTT, there are not enough studies to draw firm conclusions about their usefulness in diagnosing Q hypersensitivity or in finding safe alternatives to predict cross-reactivity.

7.4 | Drug Provocation Tests

For non-immediate QHRs, DPT with the culprit Q is indicated in cases with questionable clinical history of HR and in cases with negative allergy workup (skin testing and/or LTT) who have experienced only a mild reaction, such as delayed urticaria or MPE [2]. DPT may also be used to identify alternatives in cases where the offending Q has previously tested positive either in vivo or in vitro and also for those who have had a SCAR with the offending Q [85]. DPT with the offending Q is not recommended in patients with a history of severe and life-threatening reactions, such as SJS, TEN, interstitial nephritis, hepatitis, or hemolytic anemia [84].

The optimal protocol is a topic of debate. Most published protocols recommend reaching the full dose within 2 days, starting at 1/100th of the dose on the first day and increasing to 1/4 of the target dose, with the remaining ¾ administered on the second day [2, 22] (Table 4). In NIHRs, full therapeutic doses are used for several days thereafter, but there is no consensus on this. Prolonged DPT regimens have higher negative predictive values than shorter ones, although they are associated with more adverse effects and a greater impact on health and costs. Additionally, administering subtherapeutic doses over days increases the risk of developing antibiotic resistance [68, 85].

8 | Cross-Reactivity

The available evidence on cross-reactivity among different drugs in the Q class is controversial due to the small sample sizes and the lack of well-characterized patients or patients with a confirmed diagnosis in all reported studies. Also, some studies

TABLE 4 | Drug provocation test results in immediate and non-immediate hypersensitivity reactions to quinolones.

Authors/year of publication/ references	Number of DPT performed	Responsible quinolones	Protocol	Positive DPT with the culprit	Positive DPT with alternative
González et al. [62]	15 6 5 1 3	Quinolones Moxifloxacin Ciprofloxacin Ofloxacin Levofloxacin	Multistep doses with 30 min intervals and 45 min observation period: • Moxi: 40, 100, 200 and 400 mg • Oflox: 20, 50, 100 and 200 mg • Levo and Cipro: 50, 125, 250 and 500 mg	Immediate reaction **4/5 (80%)** Moxi 3/3 (100%) Cipro 1/2 (50%)	Immediate reaction **6/10 (60%)** Moxi 3/3 (100%) Cipro 1/3 (33.3%) Oflox 1/1 (100%) Levo 1/3 (33.3%)
Venturini Diaz et al. [65]	50	Quinolones Ciprofloxacin Levofloxacin Moxifloxacin Norfloxacin Ofloxacin Pipemidic acid	Multistep doses every 30 min until the usual daily dose	General: 7/44 (16%) < 1 h: 5/11 (45.45%) 1–24 h: 0/6 (0%) > 24 h 1/17 (5.9%) Atypical symptoms 0/10 (0%)	General: 1/6 (17%) < 1 h: 1/4 (25%) 1–24 h: 0/1 (0%) > 24 h: 0/1 (0%)
Seitz et al. [20]	79	Quinolones Ciprofloxacin Levofloxacin Moxifloxacin Ofloxacin	Increased stepwise to a normal dose with intervals of 1 h. • Levo/cipro: 50, 125, 250, and 500 mg • Oflox/moxi: 25, 50, 100 and 200 mg	IR: 6/51 (11.8%) NIR: 2/28 (7.1%)	NA
Chang et al. [64]	5 3 2	Quinolones Moxifloxacin Ciprofloxacin	Multistep	Moxi: 3/3 (100%)	Cipro: 0/2 (0%)
Lobera et al. [22]	25 8 8 9	Quinolones Levofloxacin Moxifloxacin Ciprofloxacin	The doses were administered at 30 min intervals as follows: • Cipro and Levo: 50, 125, 250, and 500 mg • Moxi: 40, 100, 200, and 400 mg	General: **4/4 (100%)** Levo 1/1 (100%) Moxi 1/1 (100%) Cipro 2/2 (100%)	General: **6/21 (28.6%)** Levo 1/7 (14.3%) Moxi 4/7 (57.1%) Cipro 1/7 (14.3%)
Demir et al. [67]	215 37 49 41 47 41	Quinolones Ciprofloxacin Levofloxacin Moxifloxacin Ofloxacin Gemifloxacin	Multistep as follow: • Cipro/levo: 5–50–100–150–200 mg • Moxi: 5–50–100–100–150 mg • Oflox: 5–25–50-100–200 mg • Gemi: 4–20–40–80–180 mg	**18/23 (78.3%)** Cipro: 10/13 (76.9%) Levo: 2/2 (100%) Moxi: 2/2 (100%) Oflox: 2/3 (66.7%) Gemi: 2/3 (66.7%)	**35/192 (18.3%)** Cipro: 3/24 (12.5%) Levo: 12/47 (25.5%) Moxi: 11/39 (28.2%) Oflox: 6/44 (13.6%) Gemi: 3/38 (7.9%)

(Continues)

TABLE 4 | (Continued)

Authors/year of publication/ references	Number of DPT performed	Responsible quinolones	Protocol	Positive DPT with the culprit	Positive DPT with alternative
Doña et al. [6]	58 35 14 9	Quinolones Ciprofloxacin Levofloxacin Moxifloxacin	3-test-day Day 1: placebo Day 2: multistep 60 min interval. (5, 20, 100 mg) for cipro/levo. (5, 30, 65 mg) for moxi Day 3: (125, 125, 250 mg) for cipro/levo (100, 100, 200 mg) for moxi	**34/34 (100%)** 23 IR: 16 cipro (100%) 3 levo (100%) 4 moxi (100%) 11 NIR: 5 cipro (100%) 2 levo (100%) 4 moxi (100%)	**13/24 (54.2%)** 11 IR 2 NIR General: 8/14 cipro (57.1%) 5/9 levo (55.5%) 0/1 moxi (0%)
Loli-Ausejo et al. [75]	7	Quinolones Levofloxacin Ciprofloxacin		General: 1/3 (33.3%) Levo: 1/1 (100%) Cipro: 0/2 (0%)	General 0/4 (0%)
Chiriac et al. [69]	26	Quinolones Ciprofloxacin Levofloxacin Moxifloxacin Ofloxacin Lomefloxacin Pefloxacin Norfloxacin	Multistep protocol 30 min incremental doses 90 -min observation period • Cipro/levo: 1–5–10–50–100-250–500 mg • Moxi/oflox/lome: 2–10–50–100–200 mg • Peflox/norflox: 2–20–100–200–400 mg *If anaphylaxis: 1/10 of initial dose	Anaphylaxis patients **12/26 (46.1%)**	NA

Note: Bold value represents the overall values.
Abbreviations: Cipro, ciprofloxacin; Gemi, gemifloxacin; IR, immediate reactions; Levo, levofloxacin; Lome, lomefloxacin; moxi, moxifloxacin; NA, not available; NIR, non-immediate reactions; Oflox, ofloxacin; Peflox, pefloxacin.

present data on both IHR and NIHR together, and not all Qs are tested in every patient [2, 11, 21, 22, 62–65, 105] (Table 5).

With regard to STs, there is a high degree of cross-reactivity between first and second-generation Qs [2, 65], although cross-reactivity within the second-generation Qs is not always present [2, 21, 22, 62, 65, 68]. It has been shown that there is a low degree of cross-reactivity between second and third generation Qs [2, 22, 68], and between third and fourth generation Qs [2, 21, 22, 62–64, 68, 105], with the exception of levofloxacin in patients who react to ofloxacin and norfloxacin [21, 62, 65].

Regarding in vitro studies, radioimmunoassay studies have found sIgE toward more than one Q in 63.6%–80% of cases [11, 23]. BAT studies have reported high cross-reactivity between second-generation Qs [2]; however, cross-reactivity between levofloxacin and moxifloxacin has been shown to be lower [2, 11, 68].

Cross-reactivity between different Qs appears to be related to the similarity in their structures [22, 62]. Cross-reactivity may also be due to the different metabolites produced. According to the available data, it is difficult to establish a pattern of cross-reactivity among different generations of Qs. Levofloxacin has been suggested as a safer alternative in cases of reaction to first, second, or fourth generation Qs due to the lack of cross-reactivity with other Qs [2].

When considering DPT, only 30% of patients who react to levofloxacin also react to ciprofloxacin [2, 22, 68]. Up to 40% and 50% of patients who react to ciprofloxacin and levofloxacin, respectively, also react to moxifloxacin in DPT [2, 22, 68]. However, most cases of patients who react to moxifloxacin seem to tolerate ciprofloxacin and levofloxacin [22, 62–65, 68].

8.1 | Cross-Reactivity Between Quinolones and Other Drugs

Quaternary ammonium sIgE has been detected in 53% of patients with IHR to Qs [56]; however, without any reported evidence of clinical relevance [106].

9 | Diagnostic Algorithm in IHRs

As illustrated in Figures 2 and 3, the diagnostic work-up of a suspected QHR begins with indication for Q use, clinical presentation, and the time elapsed between intake and onset of the reaction. In the case of IHR, as shown in Figure 2, a paired serum

TABLE 5 | Cross-reactivity among quinolones.

References	Culprit	Drugs tested; positive cases/cases in which the test was performed (%)							
		Pip	**Cip**	**Nor**	**Oflo**	**Lome**	**Lev**	**Moxi**	**Gemi**
Skin tests									
Doña et al. [6]	Pip	ND	2/2 (SPT)	ND	ND	ND	1/1 (SPT)	1/1 (SPT)	ND
Venturini Diaz et al. [65]		½	½	0/1	2/2	ND	0/2	1/2	ND
Global		**½ (50)**	**¾ (75)**	**0/1**	**2/2 (100)**	**NA**	**1/3 (33.33)**	**2/3 (66.66)**	**NA**
Doña et al. [6]	Cip	ND	3/18 (SPT)	ND	ND	ND	2/9 (SPT)	ND	ND
Demir et al. [68]		ND	5/30	ND	0/30	ND	4/30	0/30	0/30
Lobera et al. [22]		ND	0/5	0/5	0/5	ND	0/5	1/5	ND
Gonzalez et al. [62]		ND	1/3	0/3	1/3	ND	3/3	3/3	ND
Venturini Diaz et al. [65]		0/1	6/6	1/8	5/5	ND	2/4	4/4	ND
Global		**0/1 (0)**	**15/62 (24.19)**	**1/16 (6.25)**	**6/43 (13.95)**	**NA**	**11/51 (21.56)**	**8/42 (19.04)**	**NA**
Doña et al. [6]	Nor	ND	0/1 SPT	ND	ND	ND	0/1 SPT	ND	ND
Lobera et al. [22]		ND	0/1	0/1	1/1	ND	0/1	1/1	ND
Venturini Diaz et al. [65]		ND	0/2	1/3	3/3	ND	3/3	2/3	2/2
Global		**NA**	**0/4 (0)**	**¼ (25)**	**4/4 (100)**	**NA**	**3/5 (60)**	**¾ (75)**	**2/2 (100)**
Demir et al. [68]	Oflo	ND	0/4	ND	ND	ND	¼	0/4	0/4
Gonzalez et al. [62]		ND	0/1	0/1	0/1	ND	1/1	1/1	ND
Ben Said et al. [21]		0/2	ND	0/2	2/2	ND	½	0/2	ND
Venturini Diaz et al. [65]		0/1	2/2	0/2	2/3	ND	1/1	1/1	ND
Global		**0/3 (0)**	**2/3 (66.66)**	**0/5 (0)**	**4/6 (66.66)**	**NA**	**¾ (75)**	**2/4 (50)**	**NA**
Doña et al. [6]	Lev	ND	0/8 (SPT)	ND	ND	ND	2/8 (SPT)	4/5 (SPT)	ND
Demir et al. [68]		ND	0/3	ND	1/3	ND	1/3	0/3	0/3
Lobera et al. [22]		ND	0/5	0/5	ND	ND	2/5	0/5	1/5
Gonzalez-Mancebo et al. [105]		1/1	0/1	1/1	1/1 (SPT)	ND	1/1 (SPT)	0/1	ND
Ben Said et al. [21]		0/2	ND	0/2	0/2	ND	2/2	0/2	ND
Venturini Diaz et al. [65]		ND	ND	ND	ND	ND	ND	ND	ND

(Continues)

TABLE 5 | (Continued)

References	Culprit	Drugs tested; positive cases/cases in which the test was performed (%)							
		Pip	**Cip**	**Nor**	**Oflo**	**Lome**	**Lev**	**Moxi**	**Gemi**
Global		**1/3 (33.33)**	**0/17 (0)**	**1/8 (12.5)**	**2/6 (33.33)**	**NA**	**8/19 (42.10)**	**4/16** (25)	**1/8 (12.5)**
Doña et al. [6]	Moxi	ND	1/16 (SPT)	ND	ND	ND	0/9 SPT	7/7 (SPT)	ND
Demir et al. [68]		ND	1/12	ND	0/12	ND	1/12	6/12	0/12
Lobera et al. [22]		ND	0/2	0/2	0/2	ND	0/2	0/2	ND
Gonzalez et al. [62]		ND	2/3	0/3	2/3	ND	0/3	2/3	ND
Sanchez-Morillas et al. [63]		0/1	0/1	0/1	ND	ND	0/1	1/1	ND
Chang et al. [64]		0/2	ND	ND	ND	ND	0/2	0/1	ND
Venturini Diaz et al. [65]		ND	4/8	0/7	5/8	ND	4/8	8/9	0/1
Global		**0/3 (0)**	**8/42 (19.04)**	**0/13 (0)**	**7/25 (28)**	**NA**	**5/37 (13.51)**	**24/35 (68.57)**	**0/13 (0)**
Demir et al. [68]	Gemi	ND	0/6	ND	0/6	ND	1/6	1/6	0/6
Global		**0/3 (0)**	**8/48 (16.66)**	**0/13 (0)**	**7/31 (22.58)**	**NA**	**6/43 (13.95)**	**25/41 (60.97)**	**0/19 (0)**
DPT									
Doña et al. [6]	Cip	ND	21/21	ND	ND	ND	3/5	ND	ND
Demir et al. [68]		ND	10/13	ND	2/26	ND	9/28	10/27	2/21
Lobera et al. [22]		ND	2/2	ND	ND	ND	1/5	¼	ND
Gonzalez et al. [62]		ND	½	ND	ND	ND	½	3/3	ND
Venturini Diaz et al. [65]		ND	1/1	ND	0/1	ND	ND	1/1	ND
Global		**NA**	**35/39 (89.74)**	**NA**	**2/27 (7.4)**	**NA**	**15/40 (37.5)**	**15/35 (42.85)**	**2/21 (9.52)**
Doña et al. [6]	Norflox	ND	1/1	ND	ND	ND	1/1	ND	ND
Lobera et al. [22]		ND	0/1	ND	ND	ND	0/1	1/1	ND
Venturini Diaz et al. [65]		ND	0/1	0/1	ND	ND	ND	ND	ND
Global		**NA**	**1/3 (33.33)**	**0/1 (0)**	**NA**	**NA**	**½** (50)	**1/1** (100)	**NA**
Demir et al. [68]	Oflox	ND	0/3	ND	2/3	ND	1/3	0/4	0/4
Gonzalez et al. [62]		ND	0/1	ND	ND	ND	1/1	1/1	ND

(Continues)

TABLE 5 | (Continued)

References	Culprit	Drugs tested; positive cases/cases in which the test was performed (%)							
		Pip	Cip	Nor	Oflo	Lome	Lev	Moxi	Gemi
Global		**NA**	**0/4 (0)**	**NA**	**2/3 (66.66)**	**NA**	**2/4** (50)	**1/5** (20)	**0/4 (0)**
Doña et al. [6]	Lev	ND	2/5	ND	ND	ND	5/5	0/1	ND
Demir et al. [68]		ND	1/3	ND	2/3	ND	1/1	1/3	0/3
Lobera et al. [22]		ND	1/5	ND	ND	ND	1/1	2/2	ND
Global		**NA**	**4/13 (30.76)**	**NA**	**2/3 (66.66)**	**NA**	**7/7** (100)	**3/6** (50)	**0/3 (0)**
Doña et al. [6]	Moxi	ND	5/8	ND	ND	ND	0/2	8/8	ND
Demir et al. [68]		ND	2/12	ND	1/11	ND	2/12	3/3	1/10
Lobera et al. [22]		ND	0/2	ND	ND	ND	0/1	1/1	ND
Gonzalez et al. [62]		ND	1/3	ND	1/1	ND	0/1	3/3	ND
Sánchez-Morillas, [63]		ND	ND	ND	ND	ND	0/1	ND	ND
Chang et al. [64]		ND	0/2	ND	ND	ND	ND	3/3	ND
Venturini Diaz et al. [65]		ND	0/3	0/1	ND	ND	½	3/3	ND
Global		**NA**	**8/30 (26.66)**	**0/1 (0)**	**2/12 (16.66)**	**NA**	**3/19 (15.78)**	**21/21 (100)**	**1/10 (10)**
Demir et al. [68]	Gemi	ND	1/6	ND	¼	ND	1/6	1/6	2/3
Global		**NA**	**1/6 (16.66)**	**NA**	**¼ (25)**	**NA**	**1/6 (16.66)**	**1/6 (16.66)**	**2/3 (66.66)**
RIA									
Aranda et al. [11]	Cipro	ND	2/11	ND	ND	ND	1/11	1/11	ND
Aranda et al. [11]	Levo	ND	2/3	ND	ND	ND	2/3	1/3	ND
Aranda et al. [11]	Moxi	ND	4/24	ND	ND	ND	4/24	5/24	ND
BAT									
Doña et al. [6]	Cip	ND	20/25	5/6	2/6	4/6	9/19	12/20	ND
Demir et al. [68]		ND	0/10	ND	3/9	ND	1/9	1/9	4/8
Aranda et al. [11]		ND	5/11	ND	ND	ND	1/11	1/11	ND

(Continues)

TABLE 5 | (Continued)

References	Culprit	Drugs tested; positive cases/cases in which the test was performed (%)							
		Pip	Cip	Nor	Oflo	Lome	Lev	Moxi	Gemi
Global		**NA**	**25/46 (54.34)**	**5/6 (83.3)**	**5/15 (33.33)**	**4/6 (66.74)**	**11/39 (28.2)**	**14/40 (35)**	**4/8 (50)**
Doña et al. [6]	Norflox	ND	1/1	ND	ND	ND	0/1	0/1	ND
Global		**NA**	**1/1 (100)**	**NA**	**NA**	**NA**	**0/1 (0)**	**0/1 (0)**	**NA**
Doña et al. [6]	Oflox	ND	1/1	ND	ND	ND	0/1	1/1	ND
Demir et al. [68]		ND	0/1	ND	1/1	ND	0/1	0/1	ND
Ben Said et al. [21]		ND	ND	ND	2/2	ND	½	ND	ND
Global		**NA**	**½ (50)**	**NA**	**3/3 (100)**	**NA**	**¼ (25)**	**½ (50)**	**NA**
Doña et al. [6]	Lev	ND	8/11	1/1	ND	ND	5/11	2/10	ND
Aranda et al. [11]		ND	3/3	ND	ND	ND	2/3	2/3	ND
Global		**NA**	**11/14 (78.57)**	**1/1** (100)	**NA**	**NA**	**7/14 (50)**	**4/13 (30.76)**	**NA**
Doña et al. [6]	Moxi	ND	26/34	9/10	8/10	6/10	12/26	20/32	ND
Demir et al. [68]		ND	0/7	ND	1/7	ND	0/7	0/7	1/1
Aranda et al. [11]		ND	15/24	ND	ND	ND	6/24	10/24	ND
Global		**NA**	**41/65 (63.07)**	**9/10 (90)**	**8/10 (80)**	**6/10 (60)**	**18/57 (31.57)**	**30/63 (47.61)**	**1/1 (100)**
Demir et al. [68]	Gemi	ND	0/2	ND	½	ND	0/2	0/2	2/3
Global		**NA**	**0/2 (0)**	**NA**	**½ (50)**	**NA**	**0/2 (0)**	**0/2 (0)**	**2/3 (66.66)**

Note: Bold value represents the overall values.
Abbreviations: Cip, ciprofloxacin; Gemi, gemifloxacin; Lev, levofloxacin; Lome, lomefloxacin; Moxi, moxifloxacin; NA, not available data; ND, not done; Norflox, norfloxacin; Oflox, ofloxacin; Pip, pipemidic acid.

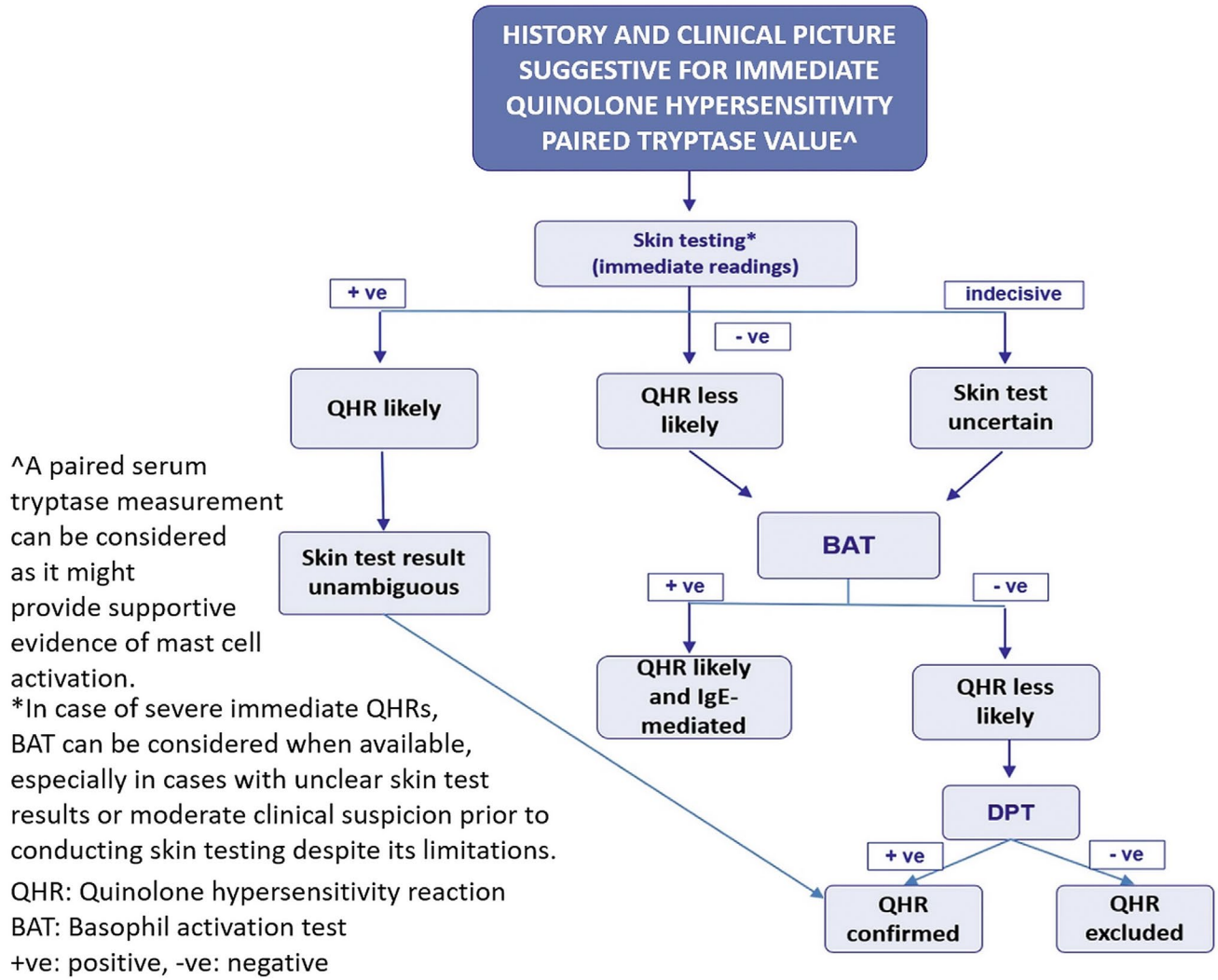


FIGURE 2 | Diagnostic algorithms for quinolone-induced immediate hypersensitivity reactions.

tryptase measurement can be considered, as it might provide supportive evidence of MCA [107]. An increased serum level of tryptase during an IHR according to the baseline confirms an IHR, whereas a normal level cannot exclude it. It is important to note that MCA is not indicative of the underlying mechanism, such as IgE-mediated or MRGPRX2-dependent mechanisms [107, 108].

In the absence of readily available and reliable drug-sIgE assays and in vitro/ex vivo tests to definitively confirm a MRGPRX2-mediated reaction, confirmation testing for immediate QHRs will typically begin with STs. However, skin testing for immediate QHRs presents unknowns and uncertainties, which require a critical and cautious interpretation of ST results. Furthermore, in the case of severe immediate QHRs (anaphylaxis) BAT can be considered when available, especially in cases with unclear skin test results or moderate clinical suspicion prior to conducting skin testing despite its limitations. Figure 3 indicates that a positive unambiguous ST result confirms a QHR without information about the mechanism. However, a doubtful positive ST response, like a negative ST result, requires complementary testing with BAT and/or DPT to confirm or exclude immediate QHRs. The utility of STs and BAT in immediate QHRs is clear. However, both strategies leave several uncertainties and weaknesses, and many patients may require DPT to document or refute the diagnosis of an immediate QHR [2] (Figure 2).

9.1 | Diagnostic Algorithm in Non-Immediate QHRs

It is not possible to create a diagnostic algorithm based on high-level evidence for NIHR due to the lack of randomized controlled trials. Our experience in this area consists mainly of case reports.

In NIHR, STs have poor sensitivity, but may be tried. Delayed reading of IDTs may be preferred in delayed-type urticaria, whereas lesional PT instead of IDT or PT on the back improves the sensitivity in individual cases of FDE [6, 20, 100, 109]. If clinical experience and resources allow, the LTT may be considered as a complementary test; although its overall sensitivity is limited, it appears to be higher in more severe reactions and when performed within a shorter time frame after the reaction [110].

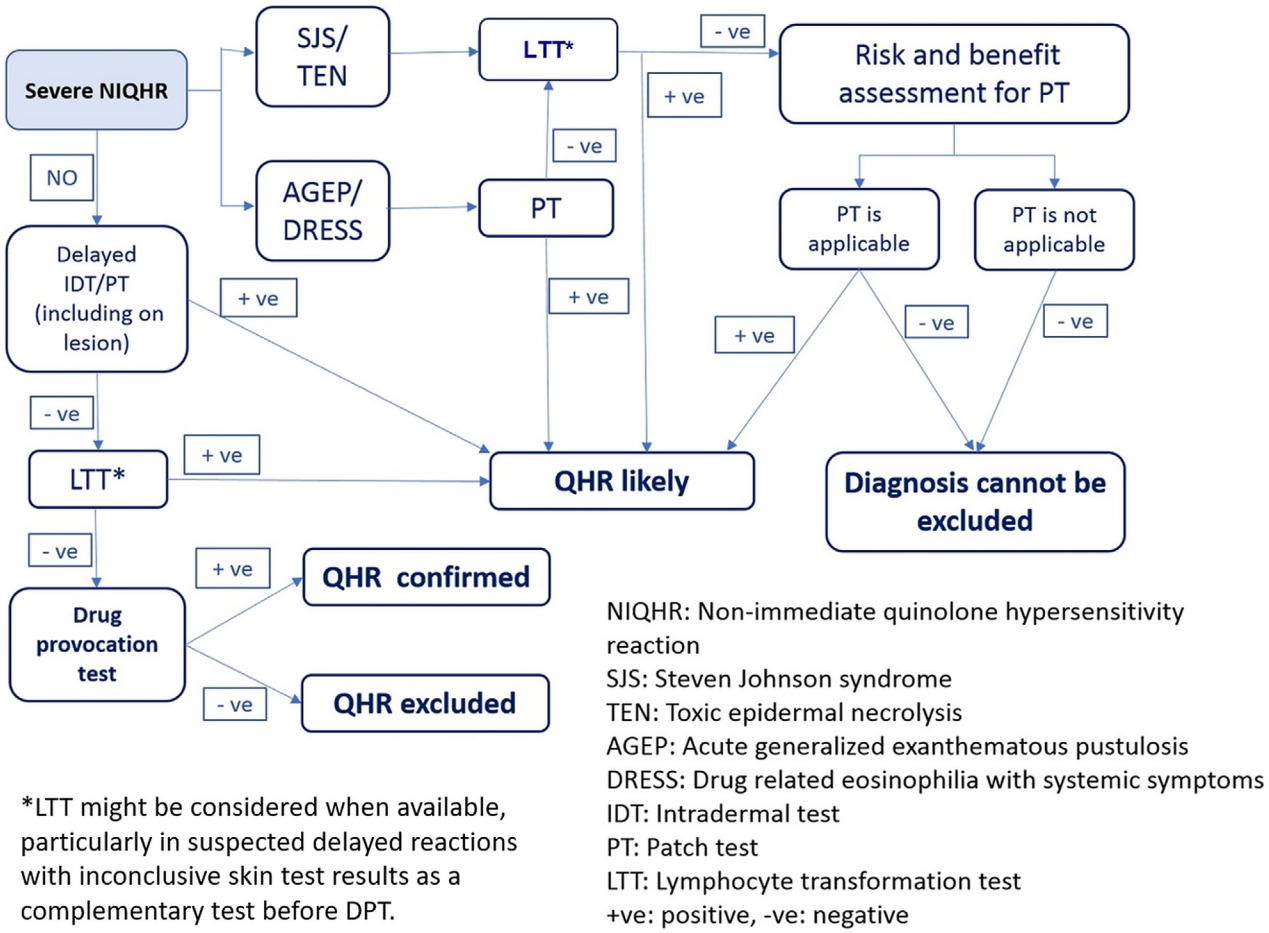


FIGURE 3 | Diagnostic algorithms for quinolone-induced nonimmediate hypersensitivity reactions.

Results of the ST or LTT should be interpreted in conjunction with the patient's history [19]. If LTT or STs yield positive results and these results are supported by clinical findings, diagnosis of non-immediate QHRs is confirmed. If LTT and STs remain negative, DPT should be performed in patients with non-severe non-immediate QHR [4, 6, 20, 63, 65, 111, 112].

PT can be performed for AGEP and DRESS as shown in the literature [19, 91, 98, 113] as an initial diagnostic test. A risk/benefit analysis should be performed in patients with SJS or TEN since there are no data available on STs for patients with SJS or TEN caused by Qs. LTT, although it has a low sensitivity, could be prioritized in the case of severe non-immediate QHRs since DPT is contraindicated in the diagnosis of severe non-immediate QHRs (Figure 3).

10 | Conclusions

In conclusion, the diagnosis of QHR is challenging, considering the complex underlying pathomechanisms involved in QHRs. In IHRs, current evidence questions the value of STs due to a high rate of false positive ST reactions, a very low positive predictive value, a lack of validated ST concentrations, and unknown sensitivity [2, 54, 55]. Differences in sensitivity of BAT in IgE-mediated IHRs do exist and could possibly be overcome by dual labeling staining for both CD63 and CD203c depending on the culprit Q. There is currently no validated assay to test MRGPRX2-dependent IHRs. Furthermore, DPTs are not standardized.

In NIHRs, evidence for STs, LTT, and DPT is even weaker. Cross-reactivity between different Qs appears to be related to the similarity in their structures. However, it is difficult to establish a pattern of cross-reactivity among different generations of Qs. We recommend two diagnostic algorithms for immediate and non-immediate QHRs. However, well-designed multicenter studies are required to make robust conclusions for values of all diagnostic tools in QHRs.

Author Contributions

All authors contributed substantially to the conception and design of the study, acquisition and interpretation of data, and drafting and revising of the manuscript. Each author has critically reviewed the final version and approved it for submission and publication.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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